

OFFSHORE / ONSHORE CRUDE OIL PIPELINE - LSAW QUALITY PLAN

Pipe Description: 36" (914.4 mm) x 15.90 mm, Grade X65MO PSL 2, LSAW (Longitudinal Submerged Arc Welded) | **Governing Standard:** IOGP S-616 / API Spec 5L 47th Ed. (Cold Expanded)

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis	One analysis per heat of steel	Ladle sample	ASTM A751	C ≤ 0.08%, P ≤ 0.015%, S ≤ 0.003%, CE_Pcm ≤ 0.19	IOGP S-616 / API 5L	H/W
2	Product Chemical Analysis	Two analyses per test unit	Pipe body coupon	ISO 14284	C ≤ 0.08%, P ≤ 0.015%, S ≤ 0.003%, CE_Pcm ≤ 0.19	API 5L Tablo 5	W
3	Pipe Body Tensile Test	Once per test unit after cold expansion	Transverse body coupon	ASTM A370	Rt0.5: 450-600 MPa, Rm: 535-760 MPa, Af ≥ 19.5%, Y/T ≤ 0.93	API 5L Tablo 7	H/W
4	Weld Seam Tensile Test	Once per test unit	Transverse weld coupon	ASTM A370	Rm ≥ 535 MPa across longitudinal weld seam	API 5L Tablo 7	W
5	CVN Charpy Impact Test (Body) at -10 °C	1 set (3 specimens) per test unit	Transverse body Charpy V	ASTM A370	At -10 °C: Min Average 50 Joules, Min Individual 38 Joules	IOGP S-616	H/W
6	CVN Charpy Impact Test (Weld & HAZ)	1 set weld + 1 set HAZ per test unit	Weld centerline and fusion line	ASTM A370	At -10 °C: Min Average 35 Joules, Min Individual 25 Joules	IOGP S-616	H/W
7	Drop Weight Tear Test (DWTT)	Once per heat at 0 °C	Transverse DWTT coupon	API RP 5L3	At 0 °C: Average shear area ≥ 85%	API 5L Tablo 18	H/W
8	Guided-Bend Test	1 root + 1 face bend per test unit	Transverse weld coupon	ASTM A370	No defect > 3.2 mm in weld or HAZ after 180° bend	API 5L 9.10	W
9	Hardness Testing	Once per test unit (13-point survey)	Cross section macro	ASTM E384	Maximum 275 HV10 (Body, Cap, Root, HAZ)	IOGP S-616	W
10	Mill Hydrostatic Pressure Test	Each pipe (100% of all pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 100% SMYS (156 bar), Duration: MINIMUM 10 SECONDS	API 5L 10.2.6.2	H/W
11	Ultrasonic Weld Seam Testing (UT)	100% full length of longitudinal weld	Full weld seam	ISO 10893-11	100% automated multi-channel UT per ISO 10893-11 Level U2	API 5L Ek E	H/W
12	Pipe Body & Ends UT Laminar Testing	100% plate scan and 100% pipe ends	Plate body & pipe ends	ISO 10893-8/9	Plate body: ISO 10893-9 Class E2; Pipe ends: 100 mm band, no defect > 6 mm	IOGP S-616	W
13	Dimensional & Wall Thickness Verification	100% of all pipes (D, t, L, Ovality)	Pipe body and pipe ends	API 5L 10.2.8	Diameter: + -0.5%; Ovality ≤ 0.6% D; Wall thickness: -8.0% / +15.0%	API 5L Tablo 10/11	W
14	Visual Surface & Marking Inspection	100% internal and external surface	Full pipe surface	API 5L 10.2.7	100% visual inspection; stencil and stamp marking verification	API 5L Madde 11	W